

MANTHAN

NEET 2027

PHYSICAL CHEMISTRY

CHEMICAL KINETICS

LECTURE -02

Subtopic – RATE & RATE OF REACTION

$$\frac{p_0 - p_s}{p_0} = x_A$$

or

$$\frac{p_0 - p_s}{p_s} = \frac{n_A}{n_B}$$

Sarvesh Sir



Today's goal



- ▶ **RATE OF REACTION**
- ▶ **MOLECULARITY**
- ▶



$$\underline{Csd} \rightarrow R.O.R. = \frac{R_{ind}}{\eta}$$

① $\eta = \text{stoichiometric coefficient}$

$$\eta = 1 \rightarrow R.O.R. = R_{ind} \text{ individual rate}$$

② $\eta > 1 \rightarrow R.O.R. < R_{ind.}$

③ $\eta < 1 \rightarrow R.O.R. > R_{ind.}$



$$r_A = \underline{0.5} \text{ mole} \cdot \text{litre}^{-1} \text{Sec}^{-1}$$

$$r_B = ?$$

$$-\frac{r_A}{1} = +\frac{r_B}{2}$$

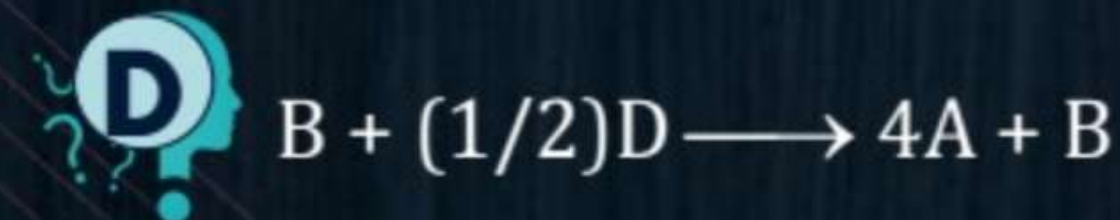
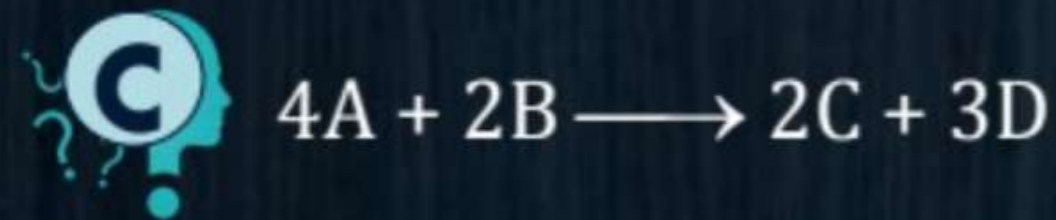
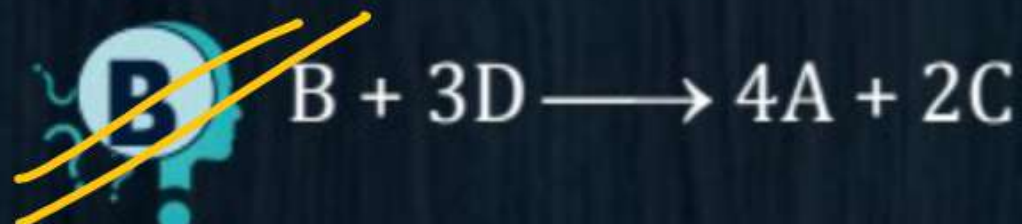
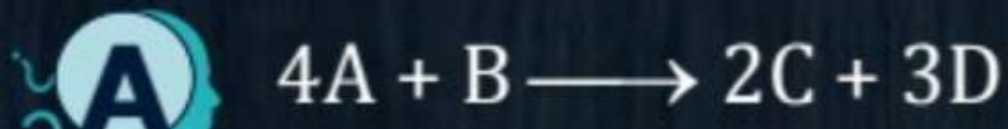
$$r_B = 2 \times r_A$$

$$r_B = \underline{2 \times 0.5}$$

Q The rate of a reaction is expressed in different ways as follows:

$$+\frac{1}{2} \frac{d[C]}{dt} = -\frac{1}{3} \frac{d[D]}{dt} = +\frac{1}{4} \frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

The reaction is:



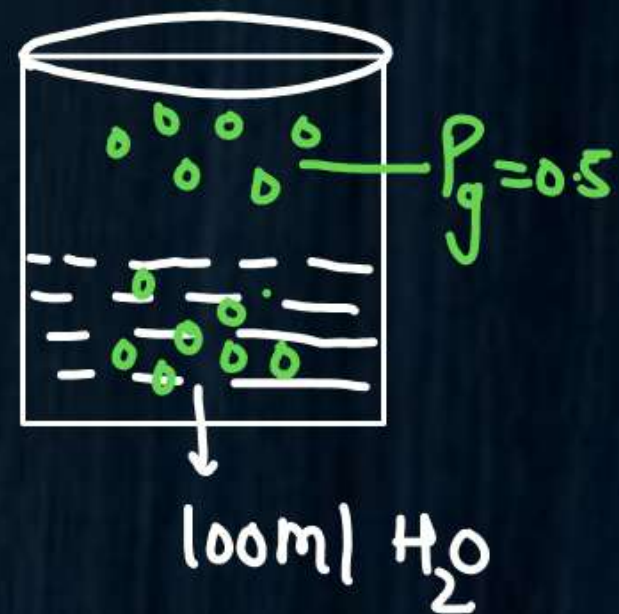
$P = 0.5 \text{ atm}$ — B.Sc. & M.Sc.
में आते हैं। or NCERT

$100 \text{ ml } H_2O$
 $(K_H)_{O_2} = 1.4 \times 10^{-3} \text{ mole.litre}^{-1} \text{ atm}^{-1}$

$$P_g \propto X_g$$

$$P_g = (K_H)_g \cdot X_g$$

$$K_H = \underline{\text{atm}}$$



→ $p_g \propto \text{Concentration (Molarity)}$

→ $C \propto p_g$

→ $C = K_H \cdot p_g$

$$\text{unit of } K_H = \frac{C}{p_g} = \frac{\text{mole/litre}}{\text{atm}}$$

$$\text{Unit of } K_H = \underline{\text{mole} \cdot \text{litre}^{-1} \text{atm}^{-1}}$$

Q. For the reaction $2A + 3B + \frac{3}{2}C \rightarrow 3P$, which statement is correct?

[JEE Main 2020- 3 September (evening)]

$$-\frac{\gamma_A}{2} = -\frac{\gamma_B}{3} = -\frac{\gamma_C}{\frac{3}{2}} = +\frac{\gamma_P}{3}$$

$$\gamma_A = \frac{2}{3}\gamma_B = \frac{4\gamma_C}{3}$$

a) $\frac{dn_A}{dt} = \frac{3}{2} \frac{dn_B}{dt} = \frac{3}{4} \frac{dn_C}{dt}$

b) $\frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{dn_C}{dt}$

~~c) $\frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{4}{3} \frac{dn_C}{dt}$~~

d) $\frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{3}{4} \frac{dn_C}{dt}$



33. Consider the chemical reaction, $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$. The rate of this reaction can be expressed in terms of time derivatives of concentration of $N_2(g)$, $H_2(g)$ or $NH_3(g)$. Identify the correct relationship amongst the rate expressions
[2002, 3M]

(a) $Rate = -\frac{d[N_2]}{dt} = -\frac{1}{3} \frac{d[H_2]}{dt} = \frac{1}{2} \frac{d[NH_3]}{dt}$

(b) $Rate = -\frac{d[N_2]}{dt} = -3 \frac{d[H_2]}{dt} = 2 \frac{d[NH_3]}{dt}$

(c) $Rate = \frac{d[N_2]}{dt} = \frac{1}{3} \frac{d[H_2]}{dt} = \frac{1}{2} \frac{d[NH_3]}{dt}$

(d) $Rate = -\frac{d[N_2]}{dt} = -\frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt}$

Solution:



Q13. For the reaction, $\text{N}_2\text{O}_5(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$.

The value of rate of disappearance of N_2O_5 is given as $6.25 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.

The rate of formation of NO_2 and O_2 is given respectively as

[CBSE AIPMT 2010]

r_{NO_2} & r_{O_2}

- (a) ~~$6.25 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$ and $6.25 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$~~
 (b) $1.25 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $3.125 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$
 (c) ~~$6.25 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$ and $3.125 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$~~
 (d) $1.25 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and ~~$6.25 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$~~

$$r_{\text{NO}_2} = 2 \times 6.25 \times 10^{-3}$$

$$r_{\text{O}_2} = \frac{6.25 \times 10^{-3}}{2}$$

$$-\frac{r_{\text{N}_2\text{O}_5}}{1} = +\frac{r_{\text{NO}_2}}{2}$$

$$r_{\text{NO}_2} = 2 \times r_{\text{N}_2\text{O}_5} = 2 \times 6.25 \times 10^{-3} \text{ mole litre}^{-1} \text{ s}^{-1}$$

Q16. In the reaction, $BrO_3^-(aq) + 5Br^-(aq) + 6H^+ \rightarrow 3Br_2(l) + 3H_2O(l)$
the rate of appearance of bromine (Br_2) is related to rate of disappearance of bromide ions as following?

$$-\frac{\gamma_{Br^-}}{5} = +\frac{\gamma_{Br_2}}{3}$$

[CBSE AIPMT 2000]

(a) $\frac{d[Br_2]}{dt} = -\frac{3}{5} \frac{d[Br^-]}{dt}$

(b) $\frac{d[Br_2]}{dt} = \frac{5}{3} \frac{d[Br^-]}{dt}$

(c) $\frac{d[Br_2]}{dt} = \frac{5}{3} \frac{d[Br^-]}{dt}$

(d) $\frac{d[Br_2]}{dt} = \frac{3}{5} \frac{d[Br^-]}{dt}$

$$\gamma_{Br_2} = -\frac{3}{5} \gamma_{Br^-}$$

Q17. For the reaction, $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, if $\frac{d[\text{NH}_3]}{dt} = 2 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$, the value of $-\frac{d[\text{H}_2]}{dt}$ would be

$$2 \times 1.5 = 3$$

$$1.5 \times 2 \times 10^{-4} = 3 \times 10^{-4}$$

[CBSE AIPMT 2009]

$$2 \times \frac{3}{2} = 3$$

~~(a) $3 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$~~

(b) $4 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

(c) $6 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

(d) $1 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

$$-\frac{\gamma_{\text{H}_2}}{3} = +\frac{\gamma_{\text{NH}_3}}{2}$$

$$-\gamma_{\text{H}_2} = \frac{3}{2} \gamma_{\text{NH}_3} = \frac{3}{2} \times 2 \times 10^{-4} = 3 \times 10^{-4}$$

Q25. Consider the reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. The equality relationship between $\frac{d[\text{NH}_3]}{dt}$ and $-\frac{d[\text{H}_2]}{dt}$ is

[CBSE AIPMT 2006]

(a) $\frac{d[\text{NH}_3]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt}$

(b) $+\frac{d[\text{NH}_3]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt}$

(c) $+\frac{d[\text{NH}_3]}{dt} = -\frac{3}{2} \frac{d[\text{H}_2]}{dt}$

(d) $\frac{d[\text{NH}_3]}{dt} = -\frac{d[\text{H}_2]}{dt}$

Q26. For the reaction, $2A + B \rightarrow 3C + D$, which of the following does not express the reaction rate?

[CBSE AIPMT 2006]

(a) $-\frac{d[C]}{3dt}$

(b) $-\frac{d[B]}{dt}$

(c) $\frac{d[D]}{dt}$

(d) $-\frac{d[A]}{2dt}$

Q35. $3A \rightarrow 2B$, rate of reaction $+\frac{d[B]}{dt}$ is equal to

$$-\frac{\gamma_A}{3} = +\frac{\gamma_B}{2}$$

[CBSE AIPMT 2002]

~~(a)~~ $-\frac{\textcircled{3}d[A]}{\textcircled{2}dt}$ $\gamma_B = -\frac{2}{3}\gamma_A$ ~~(b)~~ $-\frac{\textcircled{2}d[A]}{\textcircled{3}dt}$

(c) ~~X~~ $-\frac{1}{3}\frac{d[A]}{dt}$ (d) ~~X~~ $+2\frac{d[A]}{dt}$

Type of Rate

① Instantaneous Rate → Rate measure

in very very small interval of time

$$dt \rightarrow dc$$

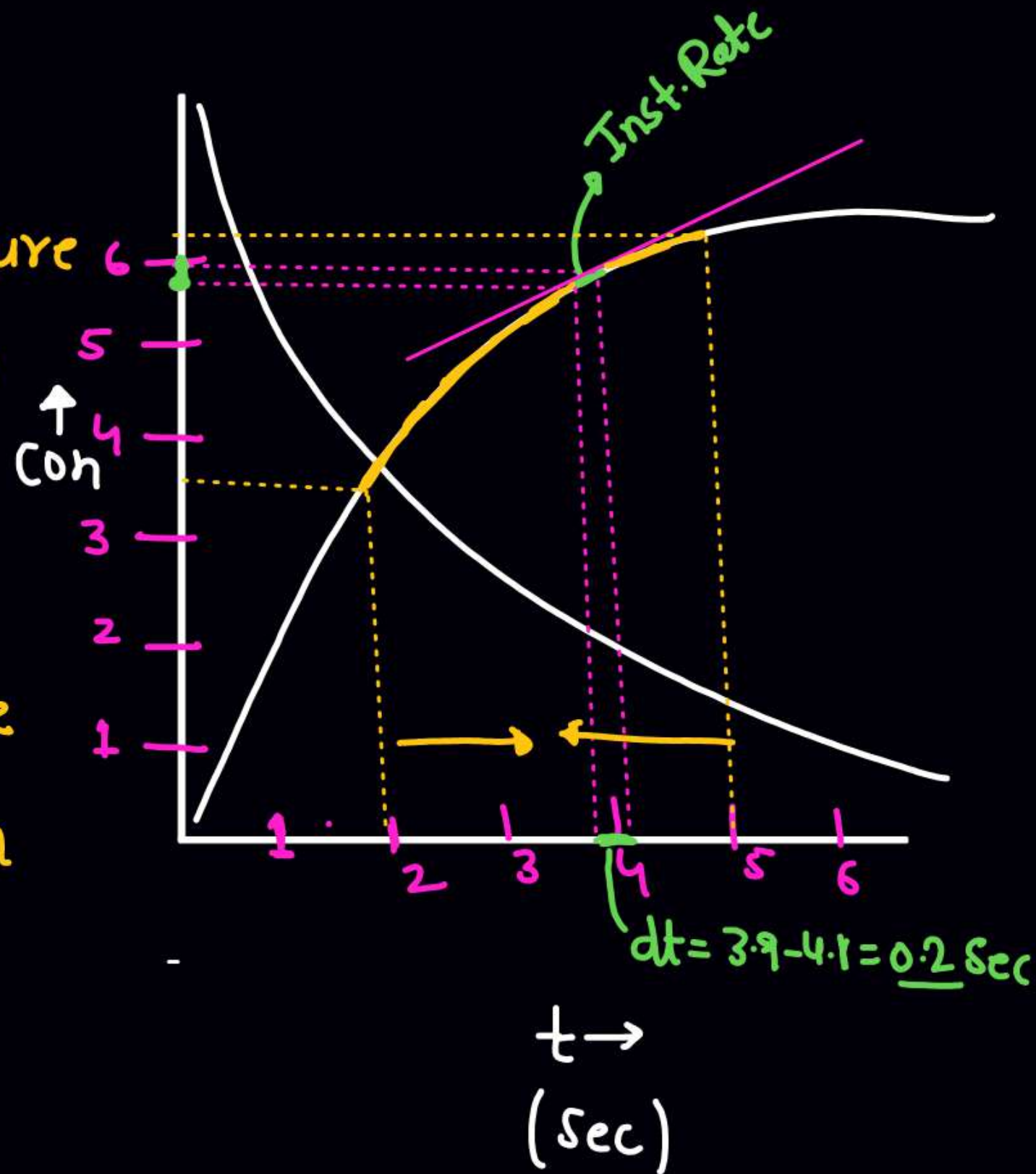
$$1\text{Sec} \rightarrow \frac{dc}{dt}$$

$$\text{Inst. Rate} \left(\frac{dc}{dt} \right) = \text{Slope of Curve}$$

② Average Rate → Rate measure in

large interval of time

$$\text{Avg. Rate} = \frac{\Delta C}{\Delta t} = \frac{\text{Change in Con.}}{\text{time taken}}$$

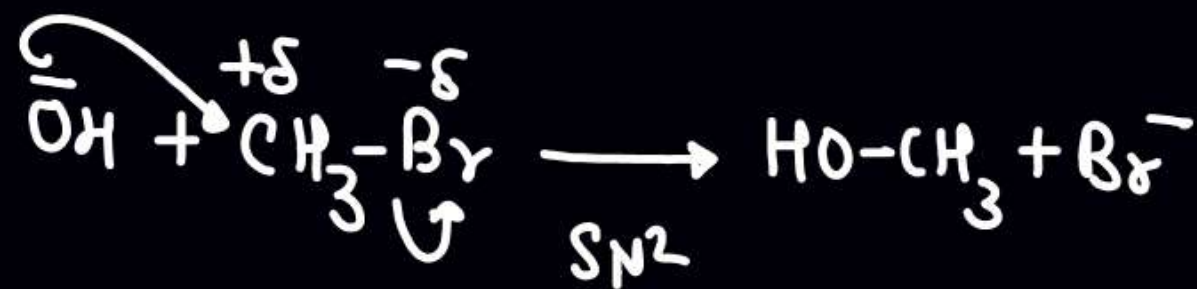


Type of Reaction in Bases of step used in whole Complete Reaction

Elementary Reaction

→ Reaction complete in single step

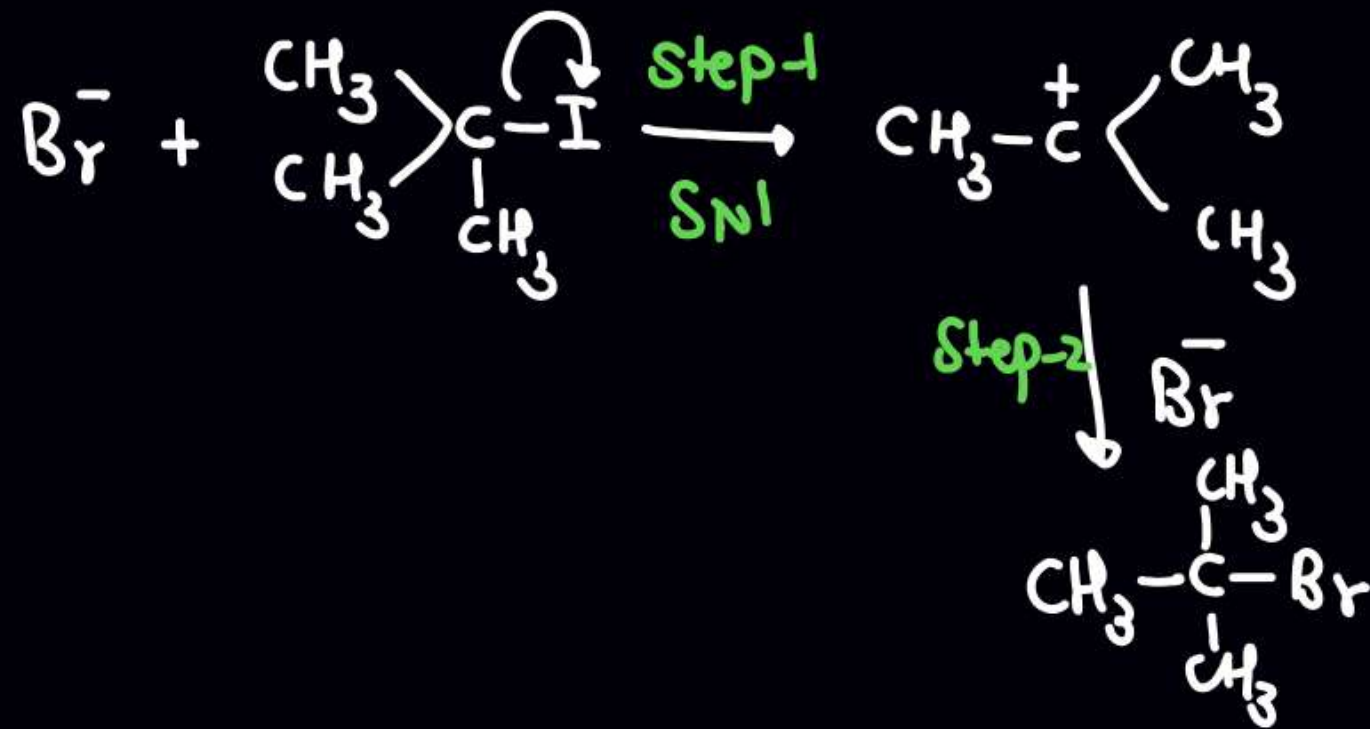
Ex → S_N^2 , E_2



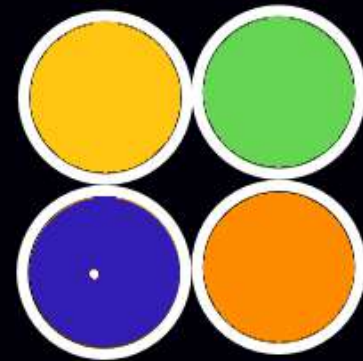
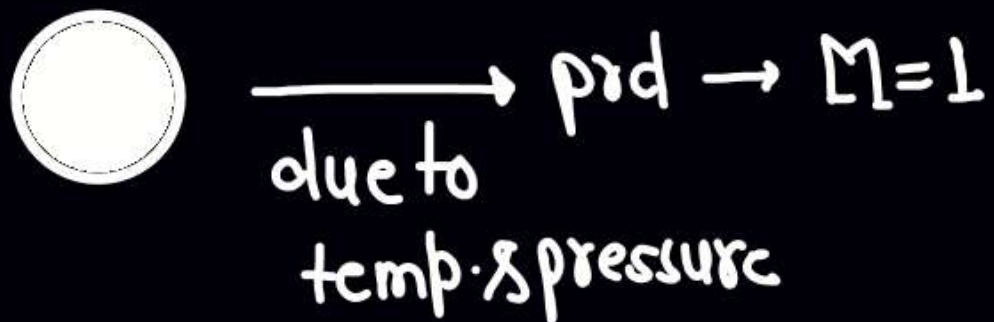
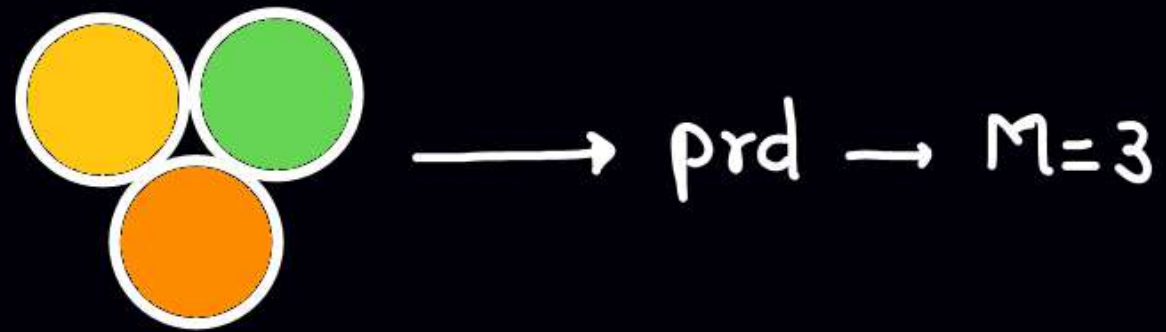
Complex Reaction

→ Reaction complete more than one step.

Ex → S_N^1 , E_1 Reaction



Molecularity $\Rightarrow$ total No. of species collides simultaneously in
a Elementary Reaction
Suppose $\rightarrow$ Reaction occurs due to collision



$\longrightarrow$ No prd form
due to energy
Reason.

$\rightarrow M = 1, 2, 3$, Max. $M = 3$

$\rightarrow M \neq 0$, fractional, Negative $\rightarrow \times$

$\rightarrow$ only define for elementary Rxn

$\rightarrow$ not define for complex rxn

$\rightarrow$ theoretical quantity

Ex $\rightarrow$ $\underline{1}A + \underline{2}B \rightarrow C \rightarrow$ Elementary rxn

$$M = 1 + 2 = \textcircled{3}$$

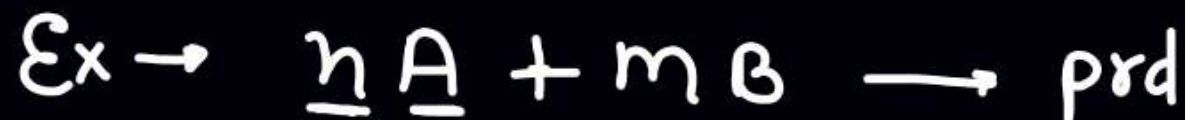
Molecularity = sum of stoichiometric of elementary rxn.

Order of Reaction → sum of power of con. of Reactant in Rate law expression Eq. of Reaction.

Collision

Rate of Reaction $\propto$ [Con. of Reactant After 't' sec]

$$\text{order of rxn} = x + y$$



→ Rate $\propto [A_t]^x [B_t]^y$ → Experimental Relation or Equation

→ $\text{Rate} = k [A_t]^x [B_t]^y$ → Rate Law expression Eq.

k = Rate Constant

↳ only depend upon nature of Rxn and temperature

$x \neq n, y \neq m$
 $x \neq n, y \neq m$

- order of rxn $\rightarrow$ Experimental quantity
- $\rightarrow$ order of rxn $\rightarrow$ may be zero, fractional, individual value may be negative
 - $\rightarrow 0, 1, 2, 3, \frac{3}{4}, \frac{1}{2}$ etc.
 - $\rightarrow$ Max. order of rxn $= 3$
 - $\rightarrow$ define elementary & complex both type of Reaction

Note $\rightarrow$ For elementary $\rightarrow$ order of rxn $=$ Molecularity
 $\rightarrow$ order of rxn theoretically value होता है।

Ex $\rightarrow 2A + B \rightarrow P \rightarrow$ elementary rxn $\rightarrow$ order of rxn $= 3$

Ex $\rightarrow 2A + B \rightarrow P \rightarrow$ Complex rxn $\rightarrow$ order of rxn $=$ not calculate
 reason $\rightarrow$ it is a Exp. quantity

CHEMICAL KINETICS

Qsd-SHEET-2

Q-1 $2A \rightarrow B$ is an example of elementary reaction

a) Molecularity = 2

b) order of rxn = 2

c) order of rxn can not calculate, because it is an experimental quantity

d) A & B both correct

Q-2 $2A \rightarrow B$ is an example of a complex reaction

a) Molecularity = 2

b) order of rxn = 2

c) order of rxn can not be calculated, because it is an experimental quantity

d) Molecularity of the above reaction is not defined because molecularity is only defined for elementary rxns

e) c & d both are correct

Q-3. $2A + B \rightarrow p$, the rate law expression of above reaction is $\text{Rate} = k [A] [B]^{\frac{1}{2}}$.

a) order of rxn is 3

b) order of rxn is $3/2$

c) order of rxn is 1

d) order of rxn is $\frac{1}{2}$

Q-4 $2A + 3B \rightarrow 5C$, the rate of disappearance of B is $-9.3 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$. than Calculate Rate of reaction

- a) $9.3 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$
- b) $3.1 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$
- c) $27.9 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$
- d) NONE

Q-5 Select correct statements for order of rxn & molecularity of Reaction.

- a) Max. value of order of rxn & molecularity is 3
- b) Value of order of rxn & molecularity may be fractional
- c) only order of rxn may be zero
- d) Molecularity is Experimental value while order of rxn is theoretical quantity.

a) a, d b) a, c c) a, c, d d) a, b, d

CHEMICAL KINETICS

Qsd-SHEET-1

Q-1 Select Correct Representation of Rate of Reaction



$$R.O.R. = -\frac{\gamma_A}{3} = +\frac{\gamma_B}{2}$$

a) $-\frac{\gamma_A}{3} = +\frac{\gamma_B}{2}$

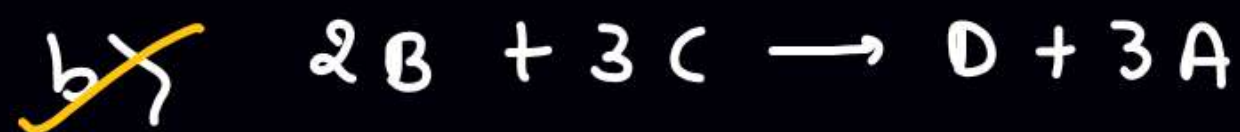
b) $-2\gamma_A = +3\gamma_B$

c) $+\frac{\gamma_B}{2} = -\frac{\gamma_A}{3}$

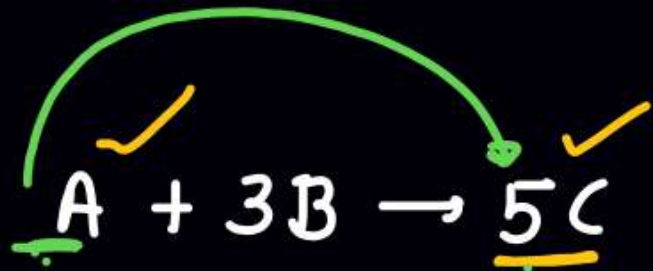
d) All Correct

Q-2 Rate of reaction expressed by

$$-\frac{\gamma_B}{\underline{2}} = \frac{\gamma_A}{\underline{3}} = \frac{\gamma_D}{\underline{1}} = -\frac{\gamma_C}{\underline{3}}$$



Q-3



gf Rate of disappearance of A is $-1.5 \times 10^3 \text{ mole} \cdot \text{litre}^{-1} \cdot \text{Sec}^{-1}$

than Calculate rate of formation of C

a) -7.5×10^3

~~b) $+7.5 \times 10^3$~~

c) 1.5×10^3

d) 0.3×10^3

$$-\frac{r_A}{1} = +\frac{r_C}{5}$$

$$r_C = 5 \times r_A$$

$$= \underline{5 \times 1.5 \times 10^3}$$

$$\underline{r_C = +7.5 \times 10^3}$$



After 10 seconds the concentration of product B form is 2.8 mole/litre. Calculate rate of disappearance of A

a) 0.28

☒ b) 0.84

c) 0.093

d) 0.14



A_0

0

$A_0 - x$

x

t=0
10 sec

$x = 2.8 \frac{\text{mole}}{\text{litre}}$

$$r_B = \frac{C_f - C_i}{t_f - t_i}$$

$$r_B = \frac{2.8 - 0}{10}$$

$$r_B = 0.28 \text{ mole.litre}^{-1}\text{sec}^{-1}$$

$$-\frac{r_A}{3} = \frac{r_B}{1}$$

$$r_A = -3r_B = -3 \times 0.28 = \underline{0.84}$$



Select correct representation of rate relation. **JEE-MAIN-2019**

$\times$ a) $\ln r_A = 2 \ln r_B$

$$-\frac{r_A}{2} = r_B$$

$\times$ b) $\ln(-r_A) = \ln(r_B)$

$$\ln\left(-\frac{r_A}{2}\right) = \ln(r_B)$$

~~c) $\ln(-r_A) = \ln B + \ln 2$~~

$$-r_A = 2 \cdot r_B$$

$\times$ d) $\ln(-r_A) + \ln 2 = \ln B$

$$\ln(-r_A) = \ln(2 \cdot r_B)$$

$$\ln(-r_A) = \ln 2 + \ln(r_B)$$